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Lab Practical #01:

Study of basic networking commands and IP configuration.

Practical Assignment #01:

1. Perform and explain various networking commands listed below:
 - i. ipconfig
 - ii. ping
 - iii. getmac
 - iv. systeminfo
 - v. traceroute / tracert
 - vi. netstat
 - vii. nslookup
 - viii. hostname
 - ix. pathping
 - x. arp

1. ipconfig

Description:

Displays all current TCP/IP network configuration values and refreshes DHCP and DNS settings.

No.	Option	Description
1	/all	Displays full configuration information.
2	/release	Releases the IP address for the specified adapter.
3	/renew	Renews the IP address for the specified adapter.
4	/flushdns	Clears the DNS Resolver cache.
5	/displaydns	Displays the contents of the DNS Resolver cache.

Implementation:



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```
C:\Users\KAVYAM>ipconfig/all
```

Windows IP Configuration

```
Host Name . . . . . : LAPTOP-58ABRMB3
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Unknown adapter Local Area Connection:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : ExpressVPN TUN Driver
Physical Address. . . . . :
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 52-5A-65-7E-F0-A1
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 52-5A-65-7E-F0-B1
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Description . . . . . : MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 50-5A-65-7E-F0-E1
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::be8b:51cd:d807:2d64%11(Preferred)
IPv4 Address. . . . . : 192.168.164.94(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 August 2025 19:16:50
Lease Expires . . . . . : 09 August 2025 20:46:57
Default Gateway . . . . . : 192.168.164.35
DHCP Server . . . . . : 192.168.164.35
DHCPv6 IAID . . . . . : 105929317
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-4D-D8-B9-BC-0F-F3-8B-27-40
DNS Servers . . . . . : 192.168.164.35
NetBIOS over Tcpip. . . . . : Enabled
```



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```
C:\Users\KAVYAM> ipconfig /release
```

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected
No operation can be performed on Ethernet while it has its media disconnected.

Unknown adapter Local Area Connection:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::be8b:51cd:d807:2d64%11  
Default Gateway . . . . . :
```

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```



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```
C:\Users\KAVYAM> ipconfig /renew
```

Windows IP Configuration

No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected
No operation can be performed on Local Area Connection* 2 while it has its media disconnected
No operation can be performed on Ethernet while it has its media disconnected.

Unknown adapter Local Area Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::be8b:51cd:d807:2d64%11
IPv4 Address. : 192.168.164.94
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.164.35

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

```
C:\Users\KAVYAM> ipconfig /flushdns
```

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.



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```
C:\Users\KAVYAM> ipconfig /displaydns
```

Windows IP Configuration

www.bing.com

```
-----  
Record Name . . . . . : www.bing.com  
Record Type . . . . . : 5  
Time To Live . . . . . : 196242  
Data Length . . . . . : 8  
Section . . . . . : Answer  
CNAME Record . . . . . : www-www.bing.com.trafficmanager.net
```

```
Record Name . . . . . : www-www.bing.com.trafficmanager.net  
Record Type . . . . . : 5  
Time To Live . . . . . : 196242  
Data Length . . . . . : 8  
Section . . . . . : Answer  
CNAME Record . . . . . : www.bing.com.edgekey.net
```

```
Record Name . . . . . : www.bing.com.edgekey.net  
Record Type . . . . . : 5  
Time To Live . . . . . : 196242  
Data Length . . . . . : 8  
Section . . . . . : Answer  
CNAME Record . . . . . : e86303.dscx.akamaiedge.net
```

```
Record Name . . . . . : e86303.dscx.akamaiedge.net  
Record Type . . . . . : 1  
Time To Live . . . . . : 196242  
Data Length . . . . . : 4  
Section . . . . . : Answer  
A (Host) Record . . . . : 184.86.248.99
```



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2. ping

Description:

Sends ICMP Echo Request messages to verify if a host is reachable over the network.

No.	Option	Description
1	-t	Pings the target until stopped manually.
2	-n count	Specifies the number of Echo Request messages to send.
3	-l size	Sends packets with a custom byte size.
4	-4	Forces using IPv4.
5	-a	Resolve addresses to hostnames.



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```
C:\Users\KAVYAM>ping -t google.com

Pinging google.com [142.251.42.238] with 32 bytes of data:
Reply from 142.251.42.238: bytes=32 time=370ms TTL=113
Reply from 142.251.42.238: bytes=32 time=280ms TTL=113
Reply from 142.251.42.238: bytes=32 time=296ms TTL=113
Reply from 142.251.42.238: bytes=32 time=189ms TTL=113
Reply from 142.251.42.238: bytes=32 time=143ms TTL=113
Reply from 142.251.42.238: bytes=32 time=209ms TTL=113
Reply from 142.251.42.238: bytes=32 time=330ms TTL=113
Reply from 142.251.42.238: bytes=32 time=586ms TTL=113
Reply from 142.251.42.238: bytes=32 time=1393ms TTL=113
Reply from 142.251.42.238: bytes=32 time=351ms TTL=113
Reply from 142.251.42.238: bytes=32 time=178ms TTL=113
Reply from 142.251.42.238: bytes=32 time=90ms TTL=113
Reply from 142.251.42.238: bytes=32 time=104ms TTL=113
Reply from 142.251.42.238: bytes=32 time=200ms TTL=113
Reply from 142.251.42.238: bytes=32 time=108ms TTL=113
Reply from 142.251.42.238: bytes=32 time=116ms TTL=113
Reply from 142.251.42.238: bytes=32 time=127ms TTL=113
Reply from 142.251.42.238: bytes=32 time=359ms TTL=113
Reply from 142.251.42.238: bytes=32 time=135ms TTL=113
Reply from 142.251.42.238: bytes=32 time=112ms TTL=113
Reply from 142.251.42.238: bytes=32 time=163ms TTL=113
Reply from 142.251.42.238: bytes=32 time=162ms TTL=113
Reply from 142.251.42.238: bytes=32 time=169ms TTL=113
Reply from 142.251.42.238: bytes=32 time=208ms TTL=113
Reply from 142.251.42.238: bytes=32 time=119ms TTL=113
Reply from 142.251.42.238: bytes=32 time=134ms TTL=113
Reply from 142.251.42.238: bytes=32 time=110ms TTL=113
Reply from 142.251.42.238: bytes=32 time=126ms TTL=113
Reply from 142.251.42.238: bytes=32 time=171ms TTL=113
Reply from 142.251.42.238: bytes=32 time=129ms TTL=113
Reply from 142.251.42.238: bytes=32 time=144ms TTL=113

Ping statistics for 142.251.42.238:
    Packets: Sent = 31, Received = 31, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 90ms, Maximum = 1393ms, Average = 235ms
```



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```
C:\Users\WELCOME>ping /n 4 google.com
```

```
Pinging google.com [142.250.70.110] with 32 bytes of data:  
Reply from 142.250.70.110: bytes=32 time=23ms TTL=116  
Reply from 142.250.70.110: bytes=32 time=25ms TTL=116  
Reply from 142.250.70.110: bytes=32 time=22ms TTL=116  
Reply from 142.250.70.110: bytes=32 time=22ms TTL=116
```

```
Ping statistics for 142.250.70.110:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 22ms, Maximum = 25ms, Average = 23ms
```

```
C:\Users\KAVYAM>ping /l 5 google.com
```

```
Pinging google.com [142.251.221.238] with 5 bytes of data:  
Reply from 142.251.221.238: bytes=5 time=22ms TTL=116  
Reply from 142.251.221.238: bytes=5 time=24ms TTL=116  
Reply from 142.251.221.238: bytes=5 time=21ms TTL=116  
Reply from 142.251.221.238: bytes=5 time=23ms TTL=116
```

```
Ping statistics for 142.251.221.238:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 21ms, Maximum = 24ms, Average = 22ms
```

```
C:\Users\KAVYAM>Ping /4 google.com
```

```
Pinging google.com [142.251.221.238] with 32 bytes of data:  
Reply from 142.251.221.238: bytes=32 time=21ms TTL=116  
Reply from 142.251.221.238: bytes=32 time=23ms TTL=116  
Reply from 142.251.221.238: bytes=32 time=27ms TTL=116  
Reply from 142.251.221.238: bytes=32 time=22ms TTL=116
```

```
Ping statistics for 142.251.221.238:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 21ms, Maximum = 27ms, Average = 23ms
```




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```
C:\Users\KAVYAM>ping /a google.com

Pinging google.com [142.250.70.110] with 32 bytes of data:
Reply from 142.250.70.110: bytes=32 time=20ms TTL=116
Reply from 142.250.70.110: bytes=32 time=31ms TTL=116
Reply from 142.250.70.110: bytes=32 time=20ms TTL=116
Reply from 142.250.70.110: bytes=32 time=20ms TTL=116

Ping statistics for 142.250.70.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 31ms, Average = 22ms
```

3. getmac

Description:

Displays the MAC address of the system's network interfaces.

No.	Option	Description
1	/v	Shows verbose output with more details.
2	/fo table	Displays output in table format.
3	/fo list	Displays output in list format.
4	/fo csv	Displays output in CSV format.
5	/nh	Hides the column headers in the output.

Implementation:

```
C:\Users\KAVYAM>getmac /V

Connection Name Network Adapter Physical Address Transport Name
=====
Ethernet Realtek Gaming BC-0F-F3-8B-27-40 Media disconnected
Wi-Fi MediaTek MT7921 50-5A-65-7E-F0-E1 \Device\Tcpip_{821C635D-A41C-4EB4-98FE-84E8FCB34EB9}
Local Area Conn ExpressVPN TUN N/A Media disconnected

C:\Users\KAVYAM>getmac /fo table

Physical Address Transport Name
=====
N/A Media disconnected
50-5A-65-7E-F0-E1 \Device\Tcpip_{821C635D-A41C-4EB4-98FE-84E8FCB34EB9}
BC-0F-F3-8B-27-40 Media disconnected
```



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```
C:\Users\KAVYAM>getmac /FO list
```

```
Physical Address: BC-0F-F3-8B-27-40  
Transport Name: Media disconnected
```

```
Physical Address: 50-5A-65-7E-F0-E1  
Transport Name: \Device\Tcpip_{821C635D-A41C-4EB4-98FE-84E8FCB34EB9}
```

```
Physical Address: N/A  
Transport Name: Media disconnected
```

```
C:\Users\KAVYAM>getmac /fo csv
```

```
"Physical Address","Transport Name"
```

```
"N/A","Media disconnected"
```

```
"50-5A-65-7E-F0-E1","\Device\Tcpip_{821C635D-A41C-4EB4-98FE-84E8FCB34EB9}"  
"BC-0F-F3-8B-27-40","Media disconnected"
```

```
C:\Users\KAVYAM>getmac /NH
```

```
BC-0F-F3-8B-27-40 Media disconnected
```

```
50-5A-65-7E-F0-E1 \Device\Tcpip_{821C635D-A41C-4EB4-98FE-84E8FCB34EB9}
```

```
N/A Media disconnected
```

4. systeminfo

Description:

Provides detailed configuration information about the computer and its OS.

No.	Option	Description
1	Systeminfo	Shows all system information.
2		
3		
4		
5		



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Implementation:

```
C:\Users\KAVYAM>systeminfo

Host Name:                               LAPTOP-58ABRMB3
OS Name:                                 Microsoft Windows 11 Home Single Language
OS Version:                             10.0.26100 N/A Build 26100
OS Manufacturer:                       Microsoft Corporation
OS Configuration:                      Standalone Workstation
OS Build Type:                          Multiprocessor Free
Registered Owner:                       KAVYAM
Registered Organization:                HP
Product ID:                             00342-42646-98249-AAOEM
Original Install Date:                  03-08-2025, 13:53:06
System Boot Time:                       07-08-2025, 13:23:26
System Manufacturer:                   HP
System Model:                           Victus by HP Gaming Laptop 15-fa0xxx
System Type:                            x64-based PC
Processor(s):                           1 Processor(s) Installed.
                                          [01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2500 Mhz
BIOS Version:                           AMI F.28, 19-08-2024
Windows Directory:                     C:\WINDOWS
System Directory:                       C:\WINDOWS\system32
Boot Device:                            \Device\HarddiskVolume1
System Locale:                           en-us;English (United States)
Input Locale:                           00004009
Time Zone:                              (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:                  16,052 MB
Available Physical Memory:              6,567 MB
Virtual Memory: Max Size:               17,844 MB
Virtual Memory: Available:              6,566 MB
Virtual Memory: In Use:                 11,278 MB
Page File Location(s):                  C:\pagefile.sys
Domain:                                 WORKGROUP
Logon Server:                           \\LAPTOP-58ABRMB3
Hotfix(s):                              3 Hotfix(s) Installed.
                                          [01]: KB5056579
                                          [02]: KB5062553
                                          [03]: KB5063666
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```
Network Card(s):      3 NIC(s) Installed.
                      [01]: ExpressVPN TUN Driver
                        Connection Name: Local Area Connection
                        Status:          Media disconnected
                      [02]: MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
                        Connection Name: Wi-Fi
                        DHCP Enabled:    Yes
                        DHCP Server:     192.168.164.35
                        IP address(es)
                        [01]: 192.168.164.94
                        [02]: fe80::be8b:51cd:d807:2d64
                      [03]: Realtek Gaming GbE Family Controller
                        Connection Name: Ethernet
                        Status:          Media disconnected

Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
  Base Virtualization Support
  Secure Boot
  DMA Protection
  UEFI Code Readonly
  SMM Security Mitigations 1.0
  Mode Based Execution Control
  APIC Virtualization
Services Configured:
  Hypervisor enforced Code Integrity
Services Running:
  Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:

Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```

5. traceroute / tracert

Description:

Traces the path that packets take to reach a network host.

No.	Option	Description
1	-d	Prevents resolving IPs to hostnames.
2	-h <max>	Sets the maximum number of hops.
3	-w <timeout>	Sets the timeout in milliseconds.
4	-4	Forces using IPv4.
5	-6	Forces using IPv6.

Implementation:



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```
C:\Users\KAVYAM>tracert /d google.com

Tracing route to google.com [142.251.223.142]
over a maximum of 30 hops:

  1    30 ms    3 ms    4 ms  192.168.164.35
  2    *        *        *      Request timed out.
  3    *        *        *      Request timed out.
  4    *        *        *      Request timed out.
  5   110 ms   67 ms   47 ms  192.168.227.195
  6    95 ms   69 ms   56 ms  172.28.2.20
  7    *        *        *      Request timed out.
  8    *        *        *      Request timed out.
  9    *        *        *      Request timed out.
 10   119 ms   746 ms  420 ms  173.194.123.22
 11   310 ms   82 ms   84 ms  172.253.68.93
 12   341 ms   87 ms  116 ms  142.251.54.85
 13   174 ms   245 ms  187 ms  142.251.223.142

Trace complete.
```

```
C:\Users\KAVYAM>tracert /h 10 google.com

Tracing route to google.com [142.251.42.238]
over a maximum of 10 hops:

  1     6 ms    5 ms   58 ms  192.168.164.35
  2    *        *        *      Request timed out.
  3    *        *        *      Request timed out.
  4    *        *        *      Request timed out.
  5   432 ms   66 ms   58 ms  192.168.227.194
  6   100 ms   82 ms   57 ms  172.28.2.20
  7    *        *        *      Request timed out.
  8    *        *        *      Request timed out.
  9    *        *        *      Request timed out.
 10   433 ms  407 ms   90 ms  173.194.123.22

Trace complete.
```



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```
C:\Users\KAVYAM>tracert -w 2 google.com
```

```
Tracing route to google.com [142.251.42.238]  
over a maximum of 30 hops:
```

1	177 ms	4 ms	5 ms	192.168.164.35
2	*	*	*	Request timed out.
3	*	*	*	Request timed out.
4	*	*	*	Request timed out.
5	127 ms	62 ms	47 ms	192.168.227.195
6	*	252 ms	39 ms	172.28.2.20
7	*	*	*	Request timed out.
8	*	*	*	Request timed out.
9	*	*	*	Request timed out.
10	393 ms	73 ms	*	173.194.123.22
11	182 ms	*	107 ms	172.253.69.191
12	*	159 ms	288 ms	192.178.83.220
13	*	294 ms	94 ms	192.178.252.83
14	297 ms	*	141 ms	142.251.198.2
15	*	279 ms	112 ms	192.178.110.249
16	*	328 ms	*	142.250.214.107
17	*	244 ms	101 ms	pnbomb-aw-in-f14.1e100.net [142.251.42.238]

```
Trace complete.
```

```
C:\Users\KAVYAM>tracert -4 google.com
```

```
Tracing route to google.com [142.251.223.142]  
over a maximum of 30 hops:
```

1	5 ms	5 ms	4 ms	192.168.164.35
2	*	*	*	Request timed out.
3	*	*	*	Request timed out.
4	*	*	*	Request timed out.
5	3412 ms	*	*	192.168.227.195
6	*	*	*	Request timed out.
7	*	*	*	Request timed out.
8	*	*	*	Request timed out.
9	*	*	*	Request timed out.
10	307 ms	247 ms	388 ms	173.194.123.22
11	332 ms	84 ms	76 ms	172.253.68.93
12	345 ms	187 ms	66 ms	142.251.54.85
13	98 ms	72 ms	59 ms	tzdela-be-in-f14.1e100.net [142.251.223.142]

```
Trace complete.
```



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```
C:\Users\WELCOME>tracert -6 google.com

Tracing route to google.com [2404:6800:4009:823::200e]
over a maximum of 30 hops:

  1  16 ms    9 ms    15 ms    2409:40c1:3101:6088::41
  2  47 ms    25 ms   28 ms    2405:200:5210:5:3924:110:3:307
  3  *         *       *       Request timed out.
  4  *         *       *       Request timed out.
  5  *         *       *       Request timed out.
  6  71 ms    37 ms   28 ms    2405:200:801:2e00::84
  7  *         *       *       Request timed out.
  8  *         *       *       Request timed out.
  9  122 ms   67 ms   68 ms    2405:200:802:760::8
 10  115 ms   94 ms   78 ms    2405:200:802:760::8
 11  *         *       *       Request timed out.
 12  104 ms   73 ms   77 ms    2404:6800:8281:240::1
 13  131 ms   71 ms   77 ms    2404:6800:8281:240::1
 14  *         *       *       Request timed out.
 15  *        116 ms  *       2001:4860:0:1::7976
 16  79 ms    78 ms   78 ms    2001:4860:0:1::8767
 17  108 ms   72 ms   78 ms    2001:4860:0:1::f69
 18  102 ms   78 ms   85 ms    bom12s13-in-x0e.1e100.net [2404:6800:4009:823::200e]

Trace complete.
```

6. netstat

Description:

Displays network connections, routing tables, interface stats, etc.

No.	Option	Description
1	-a	Displays all connections and listening ports.
2	-n	Displays addresses and port numbers numerically.
3	-o	Displays owning process ID associated with each connection.
4	-e	Displays Ethernet statistics.
5	-s	Displays per-protocol statistics.



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Implementation:

```
D:\Sem-5\CN\Lab_1>netstat /a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:445	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:5040	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:6646	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:49664	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:49665	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:49666	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:49667	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:49672	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:49688	LAPTOP-58ABRMB3:0	LISTENING
TCP	0.0.0.0:49689	LAPTOP-58ABRMB3:0	LISTENING
TCP	10.36.1.90:139	LAPTOP-58ABRMB3:0	LISTENING



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```
C:\Users\KAVYAM>netstat -n
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49674	127.0.0.1:49675	ESTABLISHED
TCP	127.0.0.1:49675	127.0.0.1:49674	ESTABLISHED
TCP	127.0.0.1:49676	127.0.0.1:49677	ESTABLISHED
TCP	127.0.0.1:49677	127.0.0.1:49676	ESTABLISHED
TCP	127.0.0.1:55676	127.0.0.1:55679	ESTABLISHED
TCP	127.0.0.1:55679	127.0.0.1:55676	ESTABLISHED
TCP	192.168.164.94:49420	4.213.25.241:443	ESTABLISHED
TCP	192.168.164.94:49421	4.213.25.240:443	ESTABLISHED
TCP	192.168.164.94:57244	4.213.25.241:443	ESTABLISHED
TCP	192.168.164.94:57253	163.70.140.60:5222	ESTABLISHED
TCP	192.168.164.94:57256	142.251.12.188:5228	ESTABLISHED
TCP	192.168.164.94:57340	52.168.112.67:443	TIME_WAIT
TCP	192.168.164.94:57341	20.189.173.9:443	TIME_WAIT
TCP	192.168.164.94:57345	52.104.26.53:443	ESTABLISHED
TCP	192.168.164.94:57346	13.107.139.11:443	ESTABLISHED

```
C:\Users\KAVYAM>netstat -o
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	127.0.0.1:49674	LAPTOP-58ABRMB3:49675	ESTABLISHED	1828
TCP	127.0.0.1:49675	LAPTOP-58ABRMB3:49674	ESTABLISHED	1828
TCP	127.0.0.1:49676	LAPTOP-58ABRMB3:49677	ESTABLISHED	3184
TCP	127.0.0.1:49677	LAPTOP-58ABRMB3:49676	ESTABLISHED	3184
TCP	127.0.0.1:55676	LAPTOP-58ABRMB3:55679	ESTABLISHED	10148
TCP	127.0.0.1:55679	LAPTOP-58ABRMB3:55676	ESTABLISHED	6276
TCP	192.168.164.94:49420	4.213.25.241:https	ESTABLISHED	5796
TCP	192.168.164.94:49421	4.213.25.240:https	ESTABLISHED	5796
TCP	192.168.164.94:57244	4.213.25.241:https	ESTABLISHED	14972
TCP	192.168.164.94:57253	whatsapp-cdn-shv-01-hyd1:5222	ESTABLISHED	12308
TCP	192.168.164.94:57256	se-in-f188:5228	ESTABLISHED	12308
TCP	192.168.164.94:57345	52.104.26.53:https	ESTABLISHED	14972
TCP	192.168.164.94:57346	13.107.139.11:https	ESTABLISHED	14972
TCP	192.168.164.94:57347	13.107.139.11:https	ESTABLISHED	14972
TCP	192.168.164.94:57348	20.189.173.9:https	ESTABLISHED	14972



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```
D:\Sem-5\CN\Lab_1>netstat /e  
Interface Statistics
```

	Received	Sent
Bytes	624022661	130423748
Unicast packets	3427865	1629355
Non-unicast packets	23359	798
Discards	0	0
Errors	0	0
Unknown protocols	0	



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```
C:\Users\KAVYAM>netstat -s
```

IPv4 Statistics

Packets Received	= 564427
Received Header Errors	= 59
Received Address Errors	= 14698
Datagrams Forwarded	= 0
Unknown Protocols Received	= 15
Received Packets Discarded	= 4892
Received Packets Delivered	= 1123211
Output Requests	= 650154
Routing Discards	= 0
Discarded Output Packets	= 54
Output Packet No Route	= 61
Reassembly Required	= 11506
Reassembly Successful	= 1150
Reassembly Failures	= 0
Datagrams Successfully Fragmented	= 0
Datagrams Failing Fragmentation	= 0
Fragments Created	= 0

IPv6 Statistics

Packets Received	= 269016
Received Header Errors	= 0
Received Address Errors	= 25944
Datagrams Forwarded	= 0
Unknown Protocols Received	= 3
Received Packets Discarded	= 32582
Received Packets Delivered	= 249209
Output Requests	= 146228
Routing Discards	= 0
Discarded Output Packets	= 25
Output Packet No Route	= 6
Reassembly Required	= 12
Reassembly Successful	= 4
Reassembly Failures	= 0
Datagrams Successfully Fragmented	= 0



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7. nslookup

Description:

Queries the DNS to obtain domain name or IP address mapping.

No.	Option	Description
1	<domain>	Retrieves IP address for the domain.
2	<IP>	Retrieves domain name for the given IP (reverse lookup).
3	set type=MX	Retrieves Mail Exchange records.
4	set type=NS	Retrieves Name Server records.
5	server <IP>	Specifies DNS server for the query.

Implementation:

```
C:\Users\KAVYAM>nslookup google.com 1.1.1.1
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4009:822::200e
          216.58.203.46
```

```
C:\Users\KAVYAM>nslookup google.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4009:827::200e
          142.250.192.78
```



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```
C:\Users\KAVYAM>nslookup -type=mx google.com
```

```
Server: UnKnown
```

```
Address: 192.168.164.35
```

```
Non-authoritative answer:
```

```
google.com MX preference = 10, mail exchanger = smtp.google.com
```

```
C:\Users\KAVYAM>nslookup -type=ns google.com
```

```
Server: UnKnown
```

```
Address: 192.168.164.35
```

```
Non-authoritative answer:
```

```
google.com nameserver = ns2.google.com
```

```
google.com nameserver = ns4.google.com
```

```
google.com nameserver = ns3.google.com
```

```
google.com nameserver = ns1.google.com
```

```
ns1.google.com internet address = 216.239.32.10
```

```
ns1.google.com AAAA IPv6 address = 2001:4860:4802:32::a
```

```
ns2.google.com internet address = 216.239.34.10
```

```
ns2.google.com AAAA IPv6 address = 2001:4860:4802:34::a
```

```
ns4.google.com internet address = 216.239.38.10
```

```
ns4.google.com AAAA IPv6 address = 2001:4860:4802:38::a
```

```
ns3.google.com internet address = 216.239.36.10
```

```
ns3.google.com AAAA IPv6 address = 2001:4860:4802:36::a
```

```
C:\Users\WELCOME>nslookup
```

```
Default Server: UnKnown
```

```
Address: 192.168.102.69
```

```
> server 8.8.8.8
```

```
Default Server: dns.google
```

```
Address: 8.8.8.8
```



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8. hostname

Description:

Displays the current name of the computer.

No.	Option	Description
1	hostname	Prints current hostname.
2		
3		
4		
5		

Implementation:

```
C:\Users\KAVYAM>hostname  
LAPTOP-58ABRMB3
```

9. pathping

Description:

Combines the functions of ping and tracert to provide information on packet loss at each hop.

No.	Option	Description
1	-n	Prevents resolving hostnames.
2	-h <max>	Sets maximum number of hops.
3	-g <host-list>	Specifies loose source route along host list.
4	-p <ms>	Wait time between pings.
5	-q <num>	Number of queries per hop.

Implementation:



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```
C:\Users\KAVYAM>pathping -n google.com
```

```
Tracing route to google.com [216.58.200.206]  
over a maximum of 30 hops:
```

```
 0  192.168.164.94  
 1  192.168.164.35  
 2      *          *          *
```

```
Computing statistics for 25 seconds...
```

Hop	RTT	Source to Here Lost/Sent = Pct	This Node/Link Lost/Sent = Pct	Address
0				192.168.164.94
			11/ 100 = 11%	
1	32ms	11/ 100 = 11%	0/ 100 = 0%	192.168.164.35

```
Trace complete.
```

```
C:\Users\KAVYAM>pathping -h 10 google.com
```

```
Tracing route to google.com [216.58.200.206]  
over a maximum of 10 hops:
```

```
 0  LAPTOP-58ABRMB3 [192.168.164.94]  
 1  192.168.164.35  
 2      *          *          *
```

```
Computing statistics for 25 seconds...
```

Hop	RTT	Source to Here Lost/Sent = Pct	This Node/Link Lost/Sent = Pct	Address
0				LAPTOP-58ABRMB3 [192.168.164.94]
			5/ 100 = 5%	
1	22ms	5/ 100 = 5%	0/ 100 = 0%	192.168.164.35

```
Trace complete.
```

```
C:\Users\KAVYAM>pathping -g 10.120.1.1 8.8.8.8 google.com
```

```
Tracing route to google.com [142.250.192.14]  
over a maximum of 30 hops:
```

```
 0  LAPTOP-58ABRMB3 [192.168.164.94]  
 1      *          *          *
```

```
Computing statistics for 0 seconds...
```

Hop	RTT	Source to Here Lost/Sent = Pct	This Node/Link Lost/Sent = Pct	Address
0				LAPTOP-58ABRMB3 [192.168.164.94]

```
Trace complete.
```



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```
C:\Users\KAVYAM>pathping -p 500 google.com

Tracing route to google.com [142.250.192.14]
over a maximum of 30 hops:
 0  LAPTOP-58ABRMB3 [192.168.164.94]
 1  192.168.164.35
 2  * * *
Computing statistics for 50 seconds...
          Source to Here   This Node/Link
Hop  RTT      Lost/Sent = Pct  Lost/Sent = Pct  Address
 0                                LAPTOP-58ABRMB3 [192.168.164.94]
 1  16ms       3/ 100 = 3%      3/ 100 = 3%      |
 1  16ms       3/ 100 = 3%      0/ 100 = 0%      192.168.164.35

Trace complete.

C:\Users\KAVYAM>pathping -q 3 google.com

Tracing route to google.com [142.250.192.14]
over a maximum of 30 hops:
 0  LAPTOP-58ABRMB3 [192.168.164.94]
 1  192.168.164.35
 2  * * *
Computing statistics for 0 seconds...
          Source to Here   This Node/Link
Hop  RTT      Lost/Sent = Pct  Lost/Sent = Pct  Address
 0                                LAPTOP-58ABRMB3 [192.168.164.94]
 1  44ms       0/   3 = 0%      0/   3 = 0%      |
 1  44ms       0/   3 = 0%      0/   3 = 0%      192.168.164.35

Trace complete.
```

10.arp

Description:

Displays and modifies the ARP (Address Resolution Protocol) cache.

No.	Option	Description
1	-a	Displays current ARP entries.
2	-g	Displays current ARP entries (same as -a).
3	-d <IP>	Deletes ARP entry for the specified IP address.
4	-s <IP> <MAC>	Adds a static ARP entry.
5	-v	Shows verbose output.

Implementation:



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```
C:\Users\KAVYAM>arp -a
```

```
Interface: 192.168.164.94 --- 0xb
```

Internet Address	Physical Address	Type
192.168.164.35	2a-9a-48-71-95-0e	dynamic
192.168.164.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
C:\Users\KAVYAM>arp -g
```

```
Interface: 192.168.164.94 --- 0xb
```

Internet Address	Physical Address	Type
192.168.164.35	2a-9a-48-71-95-0e	dynamic
192.168.164.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
C:\Windows\System32>arp -d 10.120.1.1
```

```
C:\Windows\System32>arp -s 192.168.1.100 00-aa-bb-cc-dd-ee
```



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S

```
C:\Users\KAVYAM>arp -v
```

Displays and modifies the IP-to-Physical address translation tables used by address resolution protocol (ARP).

```
ARP -s inet_addr eth_addr [if_addr]
```

```
ARP -d inet_addr [if_addr]
```

```
ARP -a [inet_addr] [-N if_addr] [-v]
```

-a	Displays current ARP entries by interrogating the current protocol data. If inet_addr is specified, the IP and Physical addresses for only the specified computer are displayed. If more than one network interface uses ARP, entries for each ARP table are displayed.
-g	Same as -a.
-v	Displays current ARP entries in verbose mode. All invalid entries and entries on the loop-back interface will be shown.
inet_addr	Specifies an internet address.
-N if_addr	Displays the ARP entries for the network interface specified by if_addr.
-d	Deletes the host specified by inet_addr. inet_addr may be wildcarded with * to delete all hosts.
-s	Adds the host and associates the Internet address inet_addr with the Physical address eth_addr. The Physical address is given as 6 hexadecimal bytes separated by hyphens. The entry is permanent.
eth_addr	Specifies a physical address.
if_addr	If present, this specifies the Internet address of the interface whose address translation table should be modified. If not present, the first applicable interface will be used.

Example:

```
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.  
> arp -a .... Displays the arp table.
```